

Date: 02 July 2026	Team ID: SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	Maximum Marks: 5 Marks

SOLUTION ARCHITECTURE TEMPLATE

Solution Architecture Overview

The system is divided into three key layers: Presentation (UI), Application logic, and Serialization artifacts. The flowchart below illustrates the structured sequence of components:

[Presentation Layer] User inputs attributes on home.html form -> Submits POST payload to /predict
▼ (HTTP POST Request)
[Application Layer - app.py] Flask router extracts form parameters and passes them to the Policy Override Engine
▼ (If policy override rules pass)
[Logic Layer - engineer_features() & scaler.pkl] Raw variables are transformed (e.g. INCOME_PER_MEMBER) and scaled using StandardScaler
▼ (Scaled features vector)
[Logic Layer - credit_model.pkl & threshold.pkl] Random Forest model scores the vector. If prob $\geq$ 0.2312, reject application, else approve.
▼ (HTTP Response)
[Presentation Layer - result.html] Displays stylized approval/rejection decision panels and execution logs

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	renders the client input form, static EDA charts, decision outcome banners, risk probability meters, and policy logs.	HTML5, CSS (Vanilla dark fintech styles), JavaScript
Application Server	Exposes endpoint routes, processes form fields, executes Layer 1 Policy overrides, and coordinates logic flow.	Python, Flask, Gunicorn
Logic Engine	Performs ETL, merges datasets, runs SMOTE sampling, trains estimators, fits the StandardScaler, and calibrates thresholds.	Python, Pandas, NumPy, Scikit-Learn, Imbalanced-Learn
Serialization Layer	Stores fitted parameters, scaled averages, and trained Random Forest model weights to enable rapid, stateless inference.	Python Pickle (.pkl files)